

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Organic advantages: often waste products, locally available and cheap; add humus improving structure and water retention; increase soil biota; slow release reduces leaching • Organic disadvantages: nutrient composition cannot be controlled; released slowly so needs long-term planning; bulky with high water content so expensive to transport; cannot be applied to a growing crop; odour / pathogens • Inorganic advantages: composition controlled to match crop needs; nutrients released rapidly; concentrated and easy to apply • Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process); no organic matter so humus and biota decline; some toxic to worms; high solubility so leached causing eutrophication; raw materials (phosphate rock) finite	6
01.2	• Crops grown in a circulating nutrient solution (inert medium) in greenhouses; nutrient concentrations monitored and adjusted so nutrients are never limiting; abiotic factors controlled • Roots are small (all in contact with nutrients) so more growth goes into the harvested crop; no soil-borne pathogens or weeds; crop stays fresh with roots • Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping); high level of technical knowledge; only economic for high-value crops	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
02.1	• Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met • Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall • Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them • Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers • Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species	6
02.2	• Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline • Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees	2
02.3	• Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp • Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
03.1	• Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity • Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) • Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop	3
03.2	• A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) • They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) • Many are in regions where environmental degradation / habitat loss threatens the wild plants	3
03.3	• A large population may all descend from a few closely related individuals so they are genetically similar • Similar adaptations mean all are susceptible to the same change (disease, climate) • Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes	3
03.4	• Morama bean (drought resistant, soya-like protein); yeheb tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	

Question	Marking guidance	Marks
04.1	• Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) • High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow • Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) • Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes	8
04.2	• Predators are killed under licence to protect stock, reducing wild populations of protected / valued species • Better cage designs, netting, fencing, bird scarers exclude predators without killing	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	
05.1	• B - radio waves do not travel through seawater; acoustic or satellite tags transmit position	1
05.2	• Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved • Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
06.1	• Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leafs • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients	6
06.2	<i>Any 4 from:</i> • Food: wildflower grasslands for seed-eating birds; providing food species • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) • Control of competitors: removing rhododendron so wildflowers get light • Pollination: a range of flowering plants with different flowering times so pollinators have food all season • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
07.1	• Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids • Desired characteristics introduced without linked unwanted ones; faster	2
07.2	• Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy • Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage • Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa	3
07.3	• Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria • Detection in the human food chain from crops approved only for animal feed • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use	4
07.4	• Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
08.1	• Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed • Yield mapping of different parts of a field controls fertiliser application in each area for future crops • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution	3
08.2	• Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive • They are mainly available in affluent societies; poorer farmers cannot afford the inputs	2
08.3	• Provides an income that allows producers to afford basic rights such as water, education, health care and food	1
08.4	• Nitrates are leached into water bodies used as sources of potable water • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic	2
08.5	<i>Any 2 from:</i> • Anaerobic digestion by microbes in rice padi fields • Livestock intestines (ruminant digestion) • Anaerobic decay of manure / slurry	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
09.1	• $2.7 - 2.45 = 0.25$ billion = 250 million years	1

Question	Marking guidance	Marks
09.2	• For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans • Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere	2
09.3	<i>Any 3 from:</i> • Oxygen in the atmosphere absorbed ultraviolet light • A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed • Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) • Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out	3
09.4	• Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) • The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years)	2
09.5	<i>Any 2 from:</i> • Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion • Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas • Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
10.2	• Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25